

Project Management

What is a project?

- A combination of interrelated activities which must be executed in a certain order before the entire task can be completed

Examples of Projects

- Building Construction

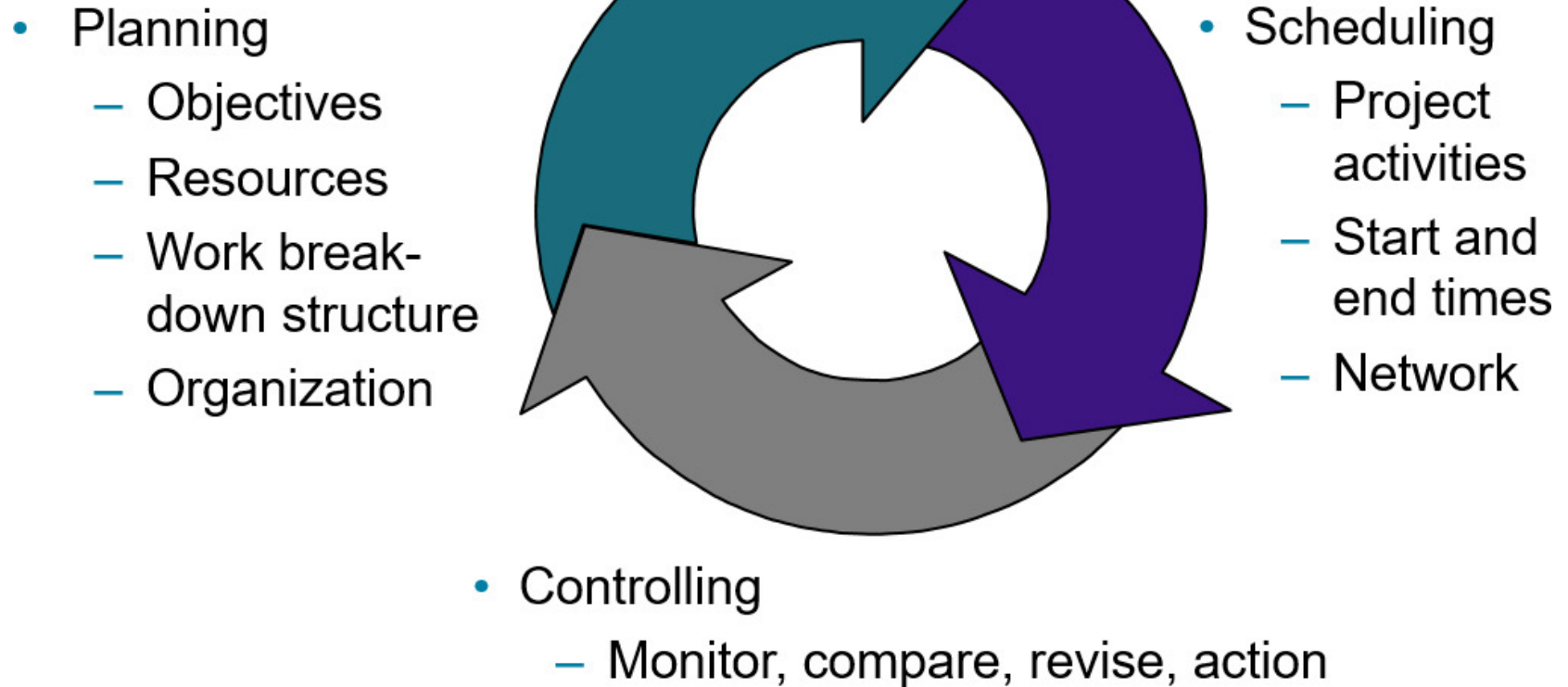


- Research Project

Management of Projects

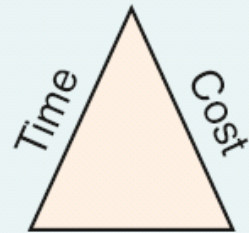
- 1. Planning** - goal setting, defining the project, team organization
- 2. Scheduling** - relate people, money, and supplies to specific activities and activities to each other
- 3. Controlling** - monitor resources, costs, quality, and budgets; revise plans and shift resources to meet time and cost demands

Project Management Activities



Project Planning, Scheduling, and Controlling

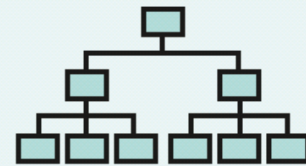
Planning the Project



Performance
Set the goals



Define the project



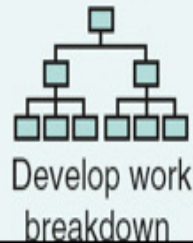
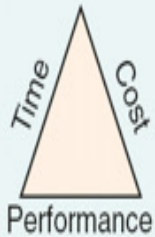
Develop work
breakdown
structure



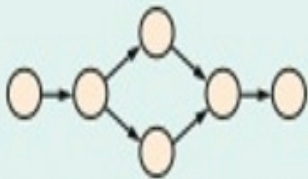
Identify team/
resources

Project Planning, Scheduling, and Controlling

Planning the Project



Scheduling the Project



Sequence activities

Adams	✓		
Smith			✓
Jones		✓	

Assign people



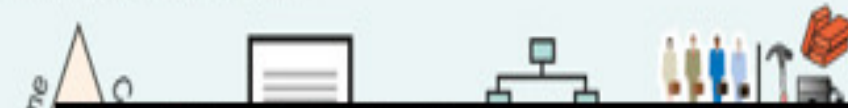
Schedule deliverables

June						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13						

Schedule resources

Project Planning, Scheduling, and Controlling

Planning the Project



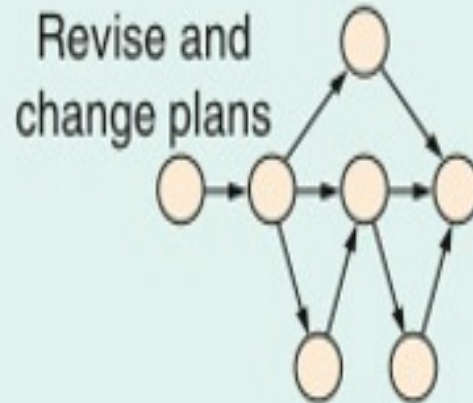
Scheduling the Project



Controlling the Project (During project)



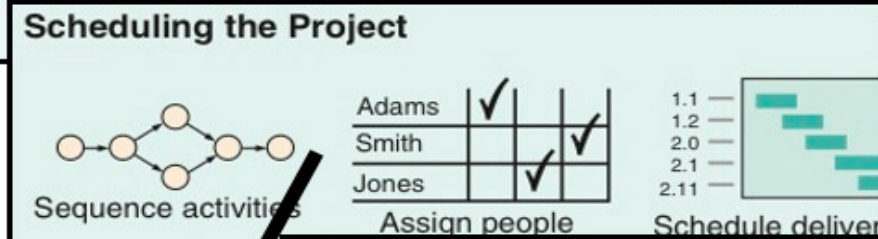
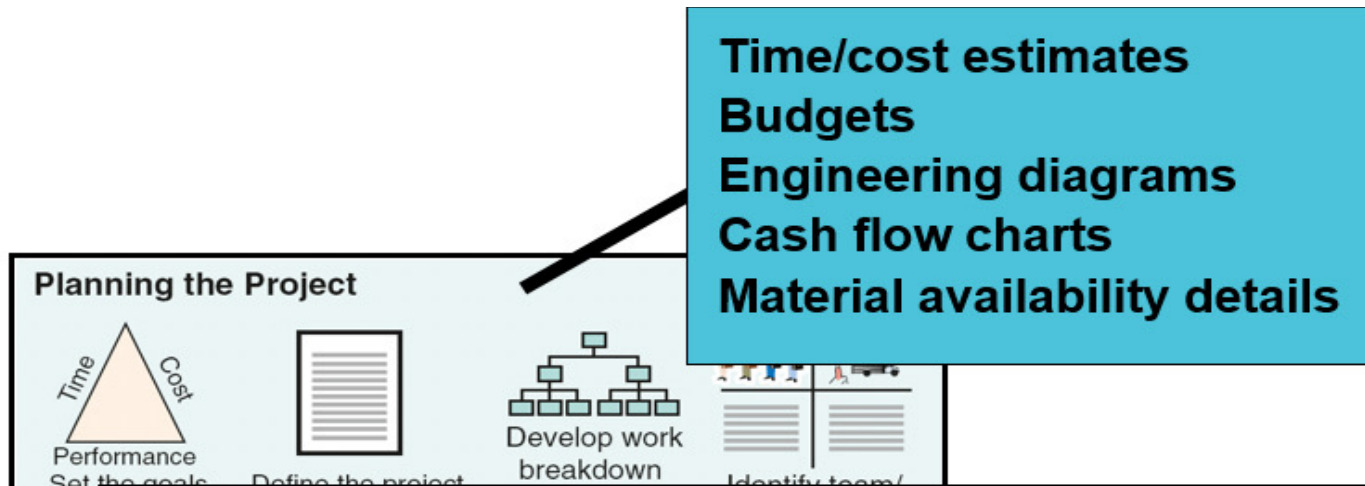
Monitor resources,
costs, quality



Adams		✓	
Smith	✓		
Jones			✓

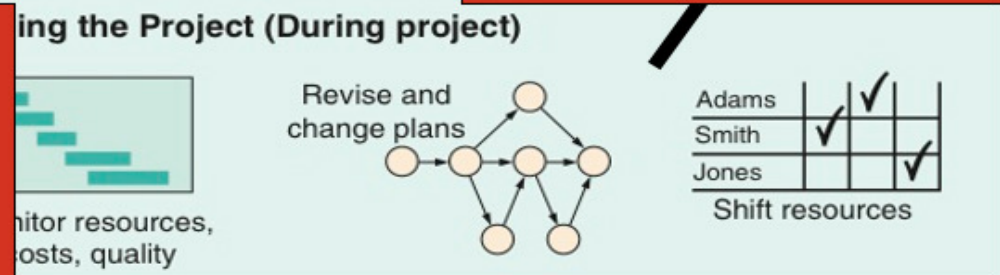
Shift resources

Project Planning, Scheduling, and Controlling



Budgets
Delayed activities report
Slack activities report

CPM/PERT
Gantt charts
Milestone charts
Cash flow schedules



The Role of the Project Manager

Highly visible

Responsible for making sure that:

- 1) All necessary activities are finished in order and on time
- 2) The project comes in within budget
- 3) The project meets quality goals
- 4) The people assigned to the project receive motivation, direction, and information

The Role of the Project Manager

Project managers should be:

- Good coaches
- Good communicators
- Able to organize activities from a variety of disciplines

Ethical Issues

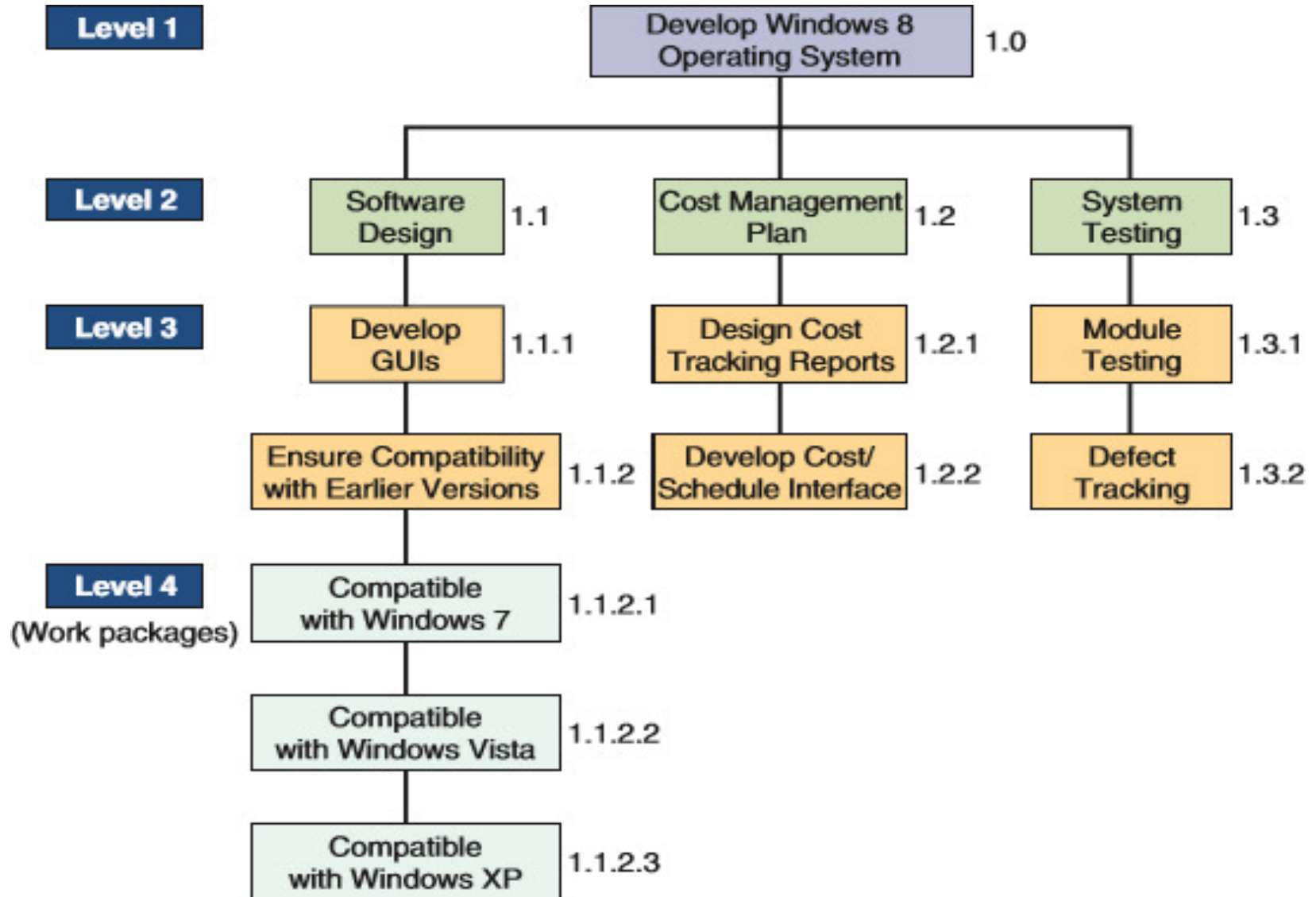
- Project managers face many ethical decisions on a daily basis
- The Project Management Institute has established an ethical code to deal with problems such as:
 - 1) Offers of gifts from contractors
 - 2) Pressure to alter status reports to mask delays
 - 3) False reports for charges of time and expenses
 - 4) Pressure to compromise quality to meet schedules

Work Breakdown Structure

Level

1. Project
2. Major tasks in the project
3. Subtasks in the major tasks
4. Activities (or “work packages”) to be completed

Work Breakdown Structure



Project Scheduling Techniques

- 1) Ensure that all activities are planned for
- 2) Their order of performance is accounted for
- 3) The activity time estimates are recorded
- 4) The overall project time is developed



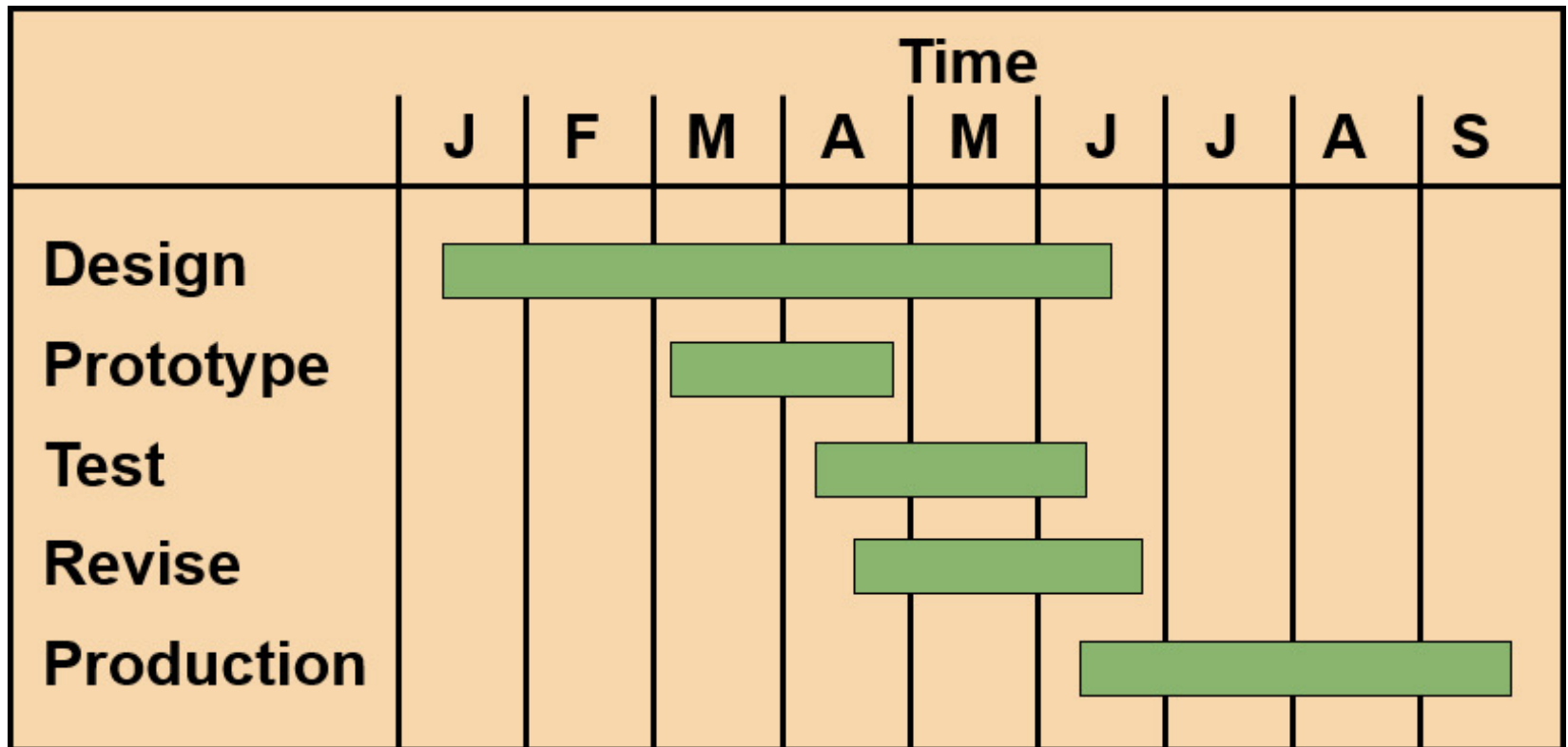
Purposes of Project Scheduling

1. Shows the relationship of each activity to others and to the whole project
2. Identifies the precedence relationships among activities
3. Encourages the setting of realistic time and cost estimates for each activity
4. Helps make better use of people, money, and material resources by identifying critical bottlenecks in the project

Project Management Techniques

- Gantt chart
- Critical Path Method (CPM)
- Program Evaluation and Review Technique (PERT)

A Simple Gantt Chart



Project Controlling

- Close monitoring of resources, costs, quality, budgets
- Feedback enables revising the project plan and shift resources
- Computerized tools produce extensive reports



Project Management Software

- There are several popular packages for managing projects
 - Oracle Primavera
 - MindView
 - HP Project
 - Fast Track
 - Microsoft Project

Project Control Reports

- 1) Detailed cost breakdowns for each task
- 2) Labor requirements
- 3) Cost and hour summaries
- 4) Raw materials and expenditure forecasts
- 5) Variance reports
- 6) Time analysis reports
- 7) Work status reports

PERT and CPM

- Network techniques
- Developed in 1950s
 - CPM by DuPont for chemical plants (1957)
 - PERT by Booz, Allen & Hamilton with the U.S. Navy, for Polaris missile (1958)
- Consider precedence relationships and interdependencies
- Each uses a different estimate of activity times

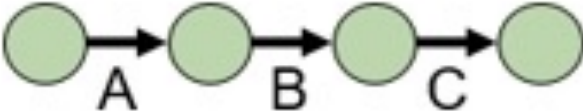
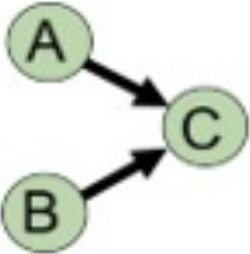
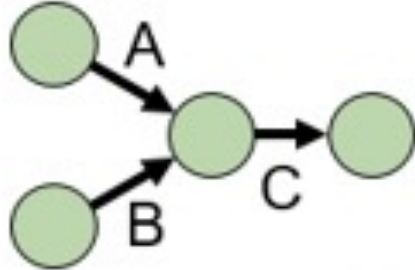
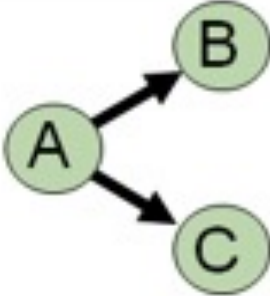
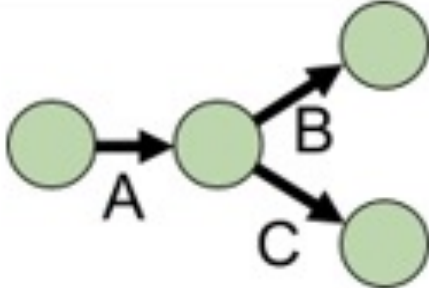
Six Steps in PERT and CPM

1. Define the project and prepare the work breakdown structure
2. Develop relationships among the activities – decide which activities must precede and which must follow others
3. Draw the network connecting all of the activities
4. Assign time and/or cost estimates to each activity
5. Compute the **longest** time path through the network – this is called the **critical path**
6. Use the network to help plan, schedule, monitor, and control the project

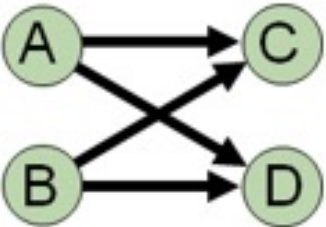
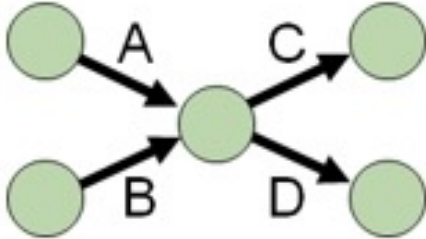
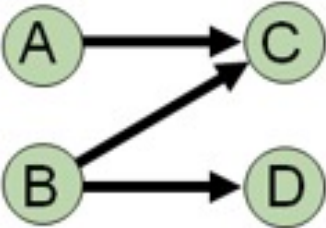
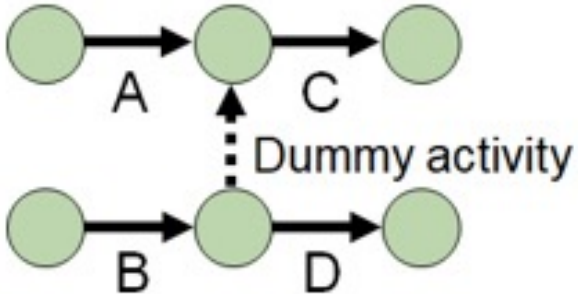
Questions PERT and CPM Can Answer

1. When will the entire project be completed?
2. What are the critical activities or tasks in the project?
3. Which are the noncritical activities?
4. What is the probability the project will be completed by a specific date?
5. Is the project on schedule, behind schedule, or ahead of schedule?
6. Is the money spent equal to, less than, or greater than the budget?
7. Are there enough resources available to finish the project on time?
8. If the project must be finished in a shorter time, what is the way to accomplish this at least cost?

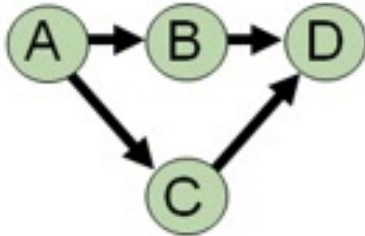
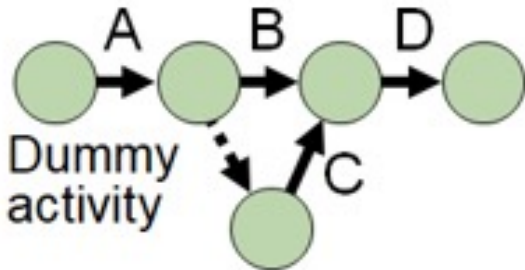
A Comparison of A O N and A O A Network Conventions

Activity On Node (AON)	Activity Meaning	Activity On Arrow (AOA)
(a) 	A comes before B, which comes before C	
(b) 	A and B must both be completed before C can start	
(c) 	B and C cannot begin until A is completed	

A Comparison of AON and AOA Network Conventions

Activity On Node (AON)	Activity Meaning	Activity On Arrow (AOA)
(d) 	C and D cannot begin until both A and B are completed	
(e) 	C cannot begin until both A and B are completed D cannot begin until B is completed A dummy activity is introduced in AOA	

A Comparison of AON and AOA Network Conventions

Activity On Node (AON)	Activity Meaning	Activity On Arrow (AOA)
(f) 	B and C cannot begin until A is completed D cannot begin until both B and C are completed A dummy activity is again introduced in AOA	

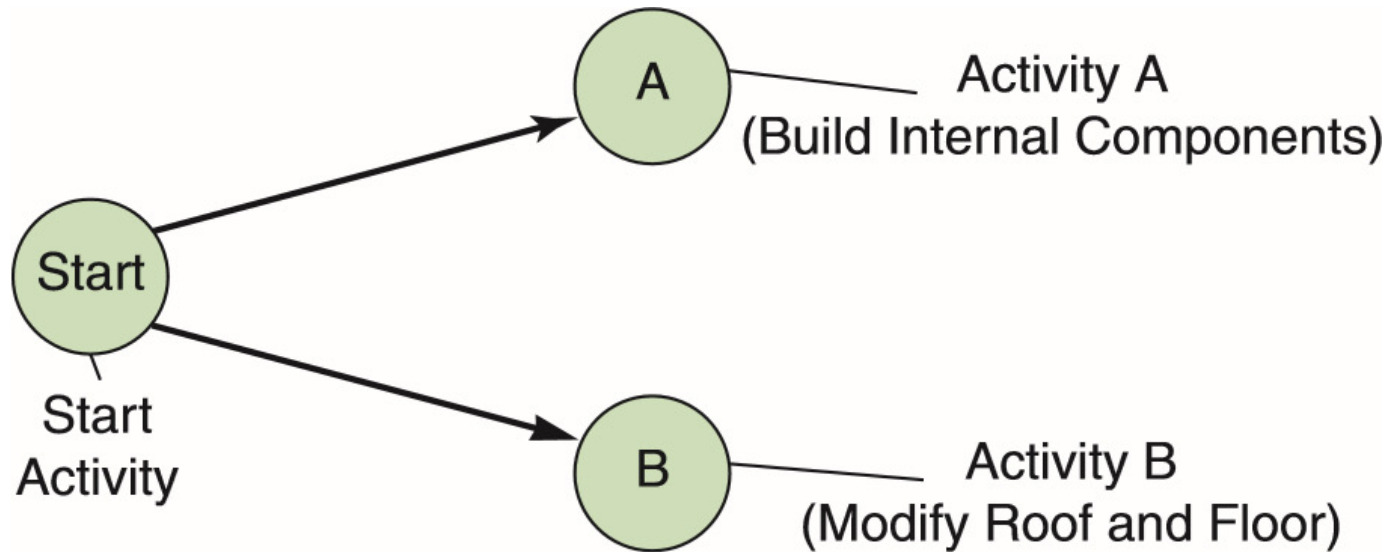
AON Example

Paper Manufacturing's Activities and Predecessors

Activity	Description	Immediate Predecessors
A	Build internal components	—
B	Modify roof and floor	—
C	Construct collection stack	A
D	Pour concrete and install frame	A, B
E	Build high-temperature burner	C
F	Install pollution control system	C
G	Install air pollution device	D, E
H	Inspect and test	E, G

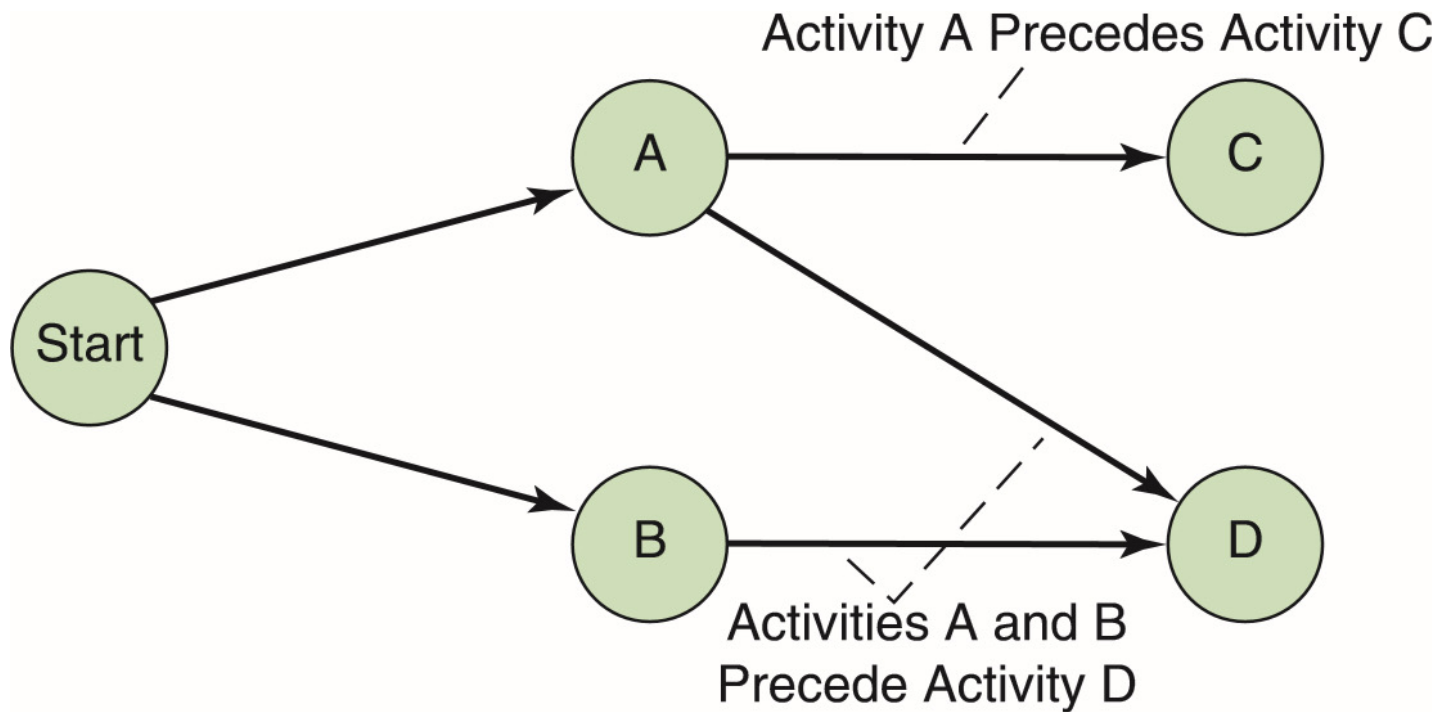
AON Network

Beginning AON Network



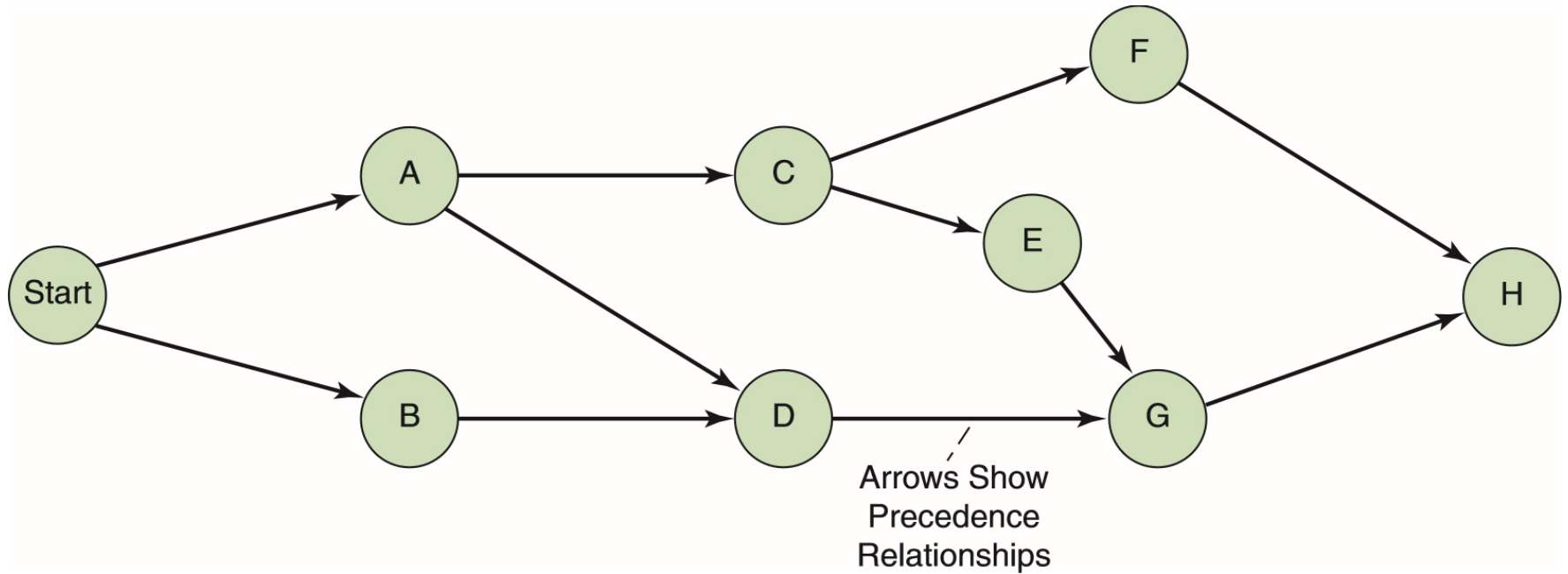
AON Network

Intermediate AON Network



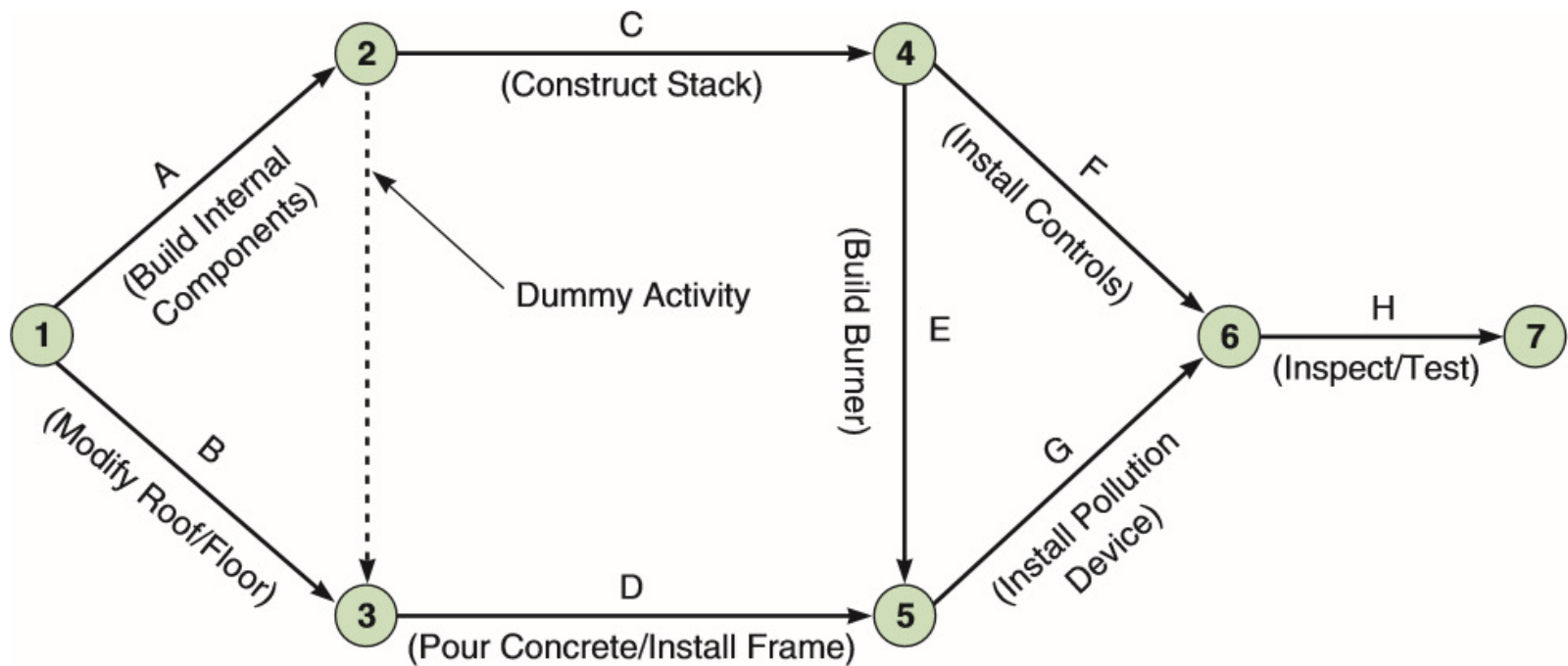
AON Network

Complete AON Network



AOA Network

Complete AOA Network (with Dummy Activity)



Determining the Project Schedule

Perform a Critical Path Analysis

- The critical path is the longest path through the network
- The critical path is the shortest time in which the project can be completed
- Any delay in critical path activities delays the project
- Critical path activities have no slack time

Determining the Project Schedule

Time Estimates of Paper Manufacturing process

Activity	Description	Time (Weeks)
A	Build internal components	2
B	Modify roof and floor	3
C	Construct collection stack	2
D	Pour concrete and install frame	4
E	Build high-temperature burner	4
F	Install pollution control system	3
G	Install air pollution device	5
H	Inspect and test	2
	Total time (weeks)	25

Determining the Project Schedule

Perform a Critical Path Analysis

Earliest start (ES) = earliest time at which an activity can start, assuming all predecessors have been completed

Earliest finish (EF) = earliest time at which an activity can be finished

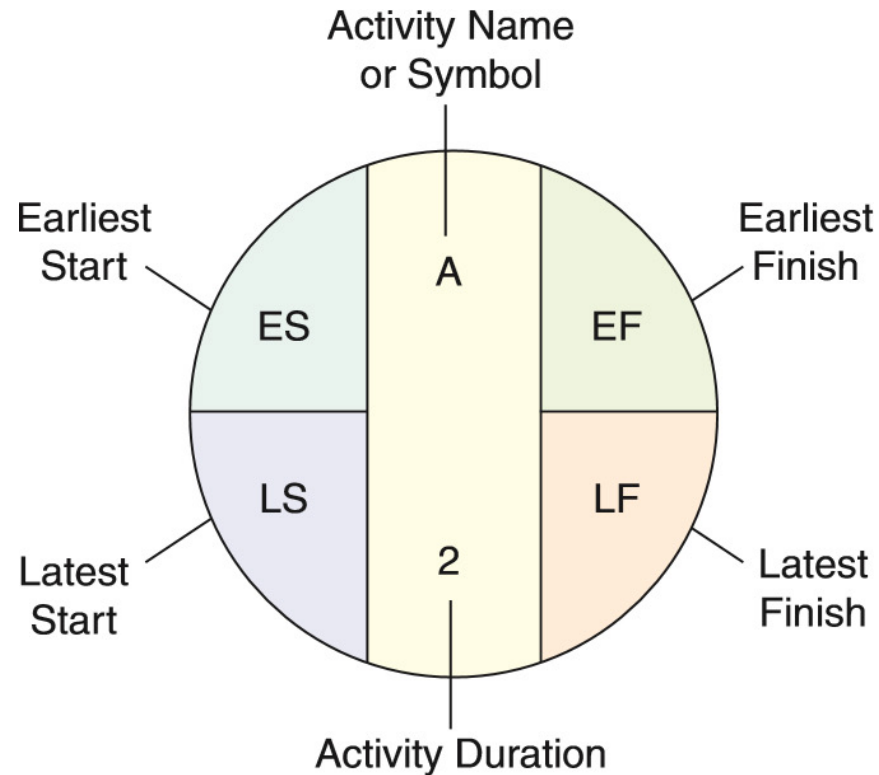
Latest start (LS) = latest time at which an activity can start so as to not delay the completion time of the entire project

Latest finish (LF) = latest time by which an activity has to be finished so as to not delay the completion time of the entire project

Determining the Project Schedule

Notation Used in Nodes for Forward and Backward Pass

Activity Format



Forward Pass

Begin at starting event and work forward

Earliest Start Time Rule:

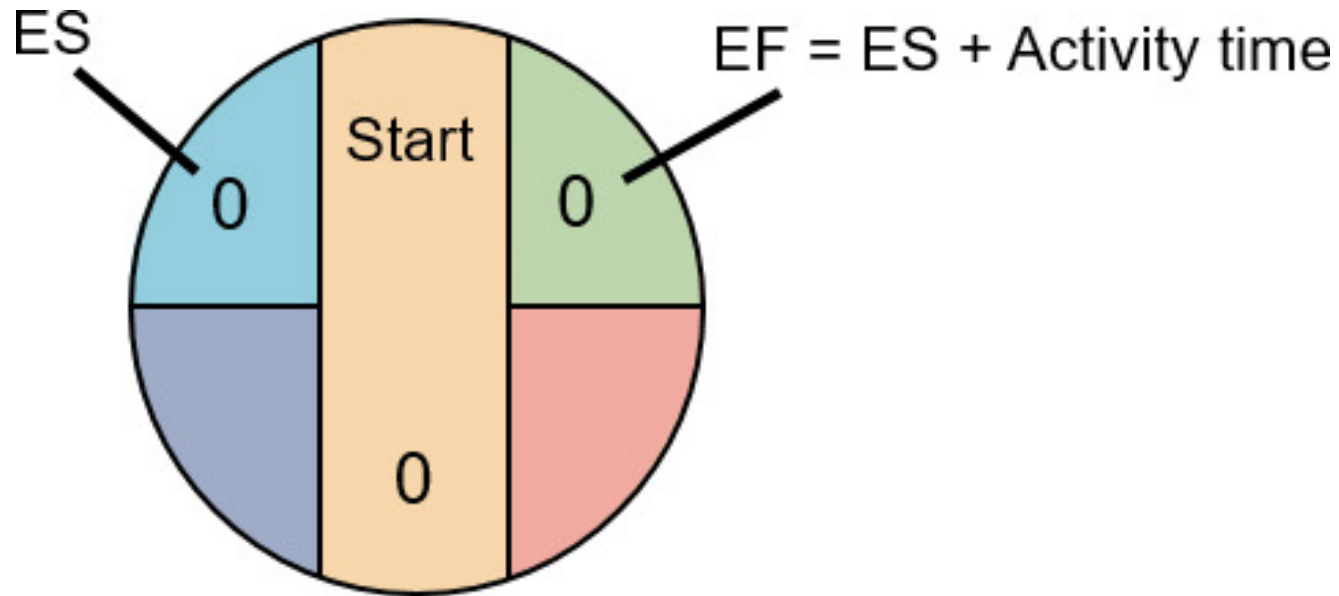
- If an activity has only a single immediate predecessor, its ES equals the EF of the predecessor
- If an activity has multiple immediate predecessors, its ES is the maximum of all the EF values of its predecessors

$$ES = \text{Max} \{EF \text{ of all immediate predecessors}\}$$

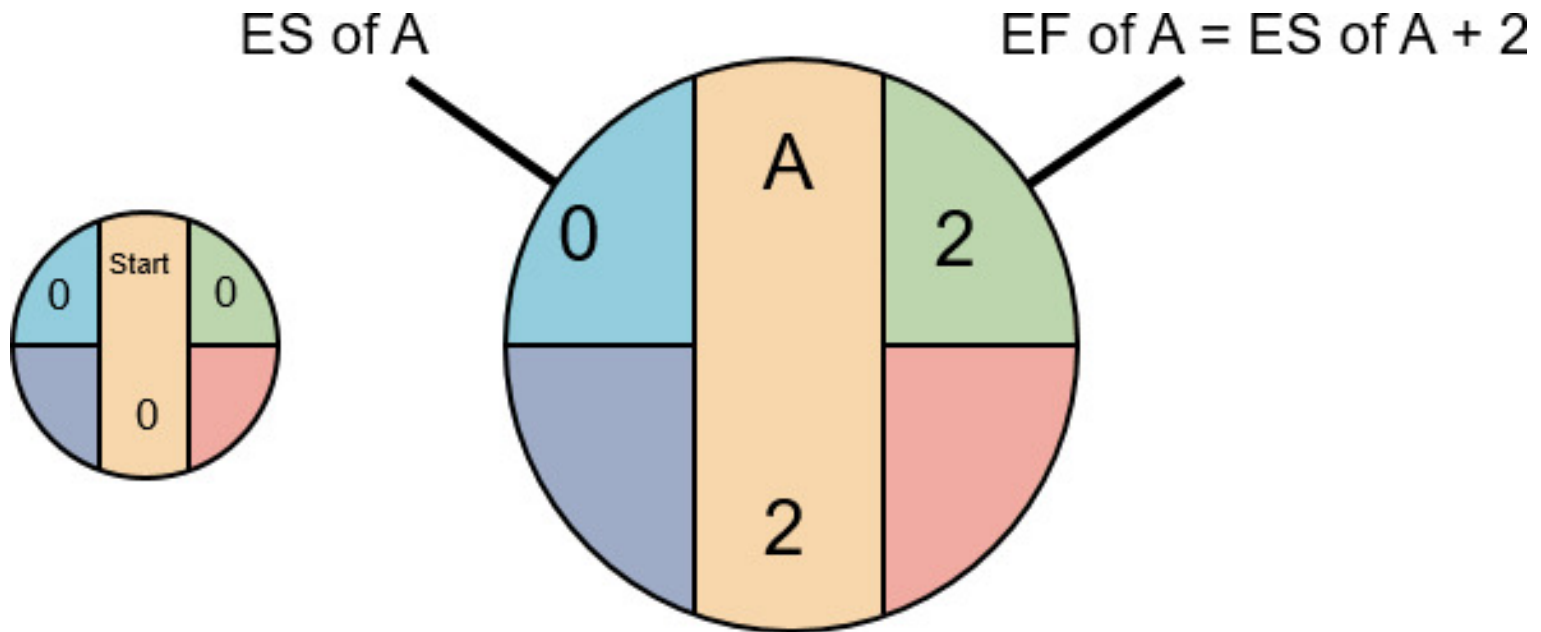
- The earliest finish time (EF) of an activity is the sum of its earliest start time (ES) and its activity time

$$EF = ES + \text{Activity time}$$

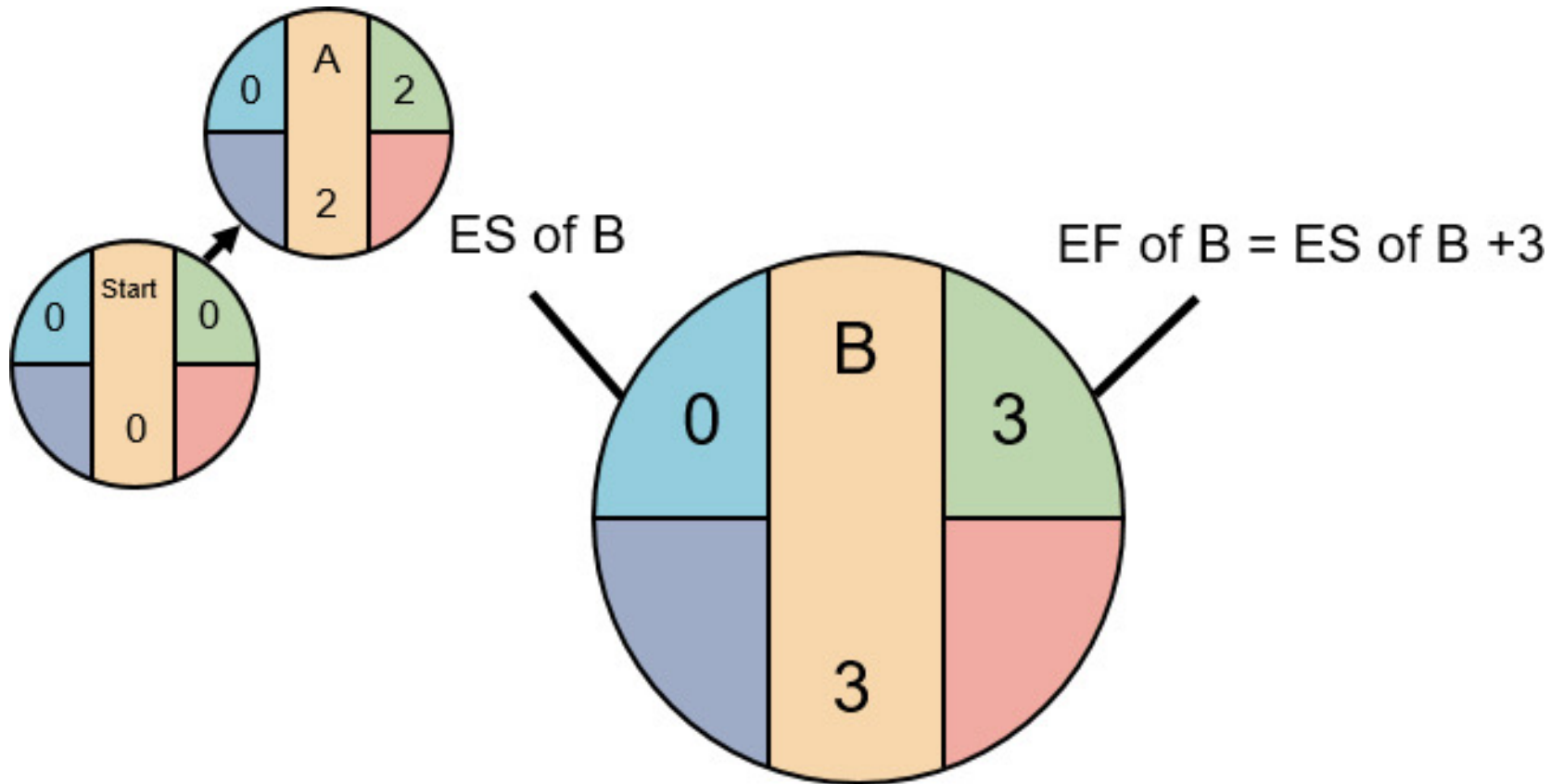
ES/EF Network for the paper manufacturing example



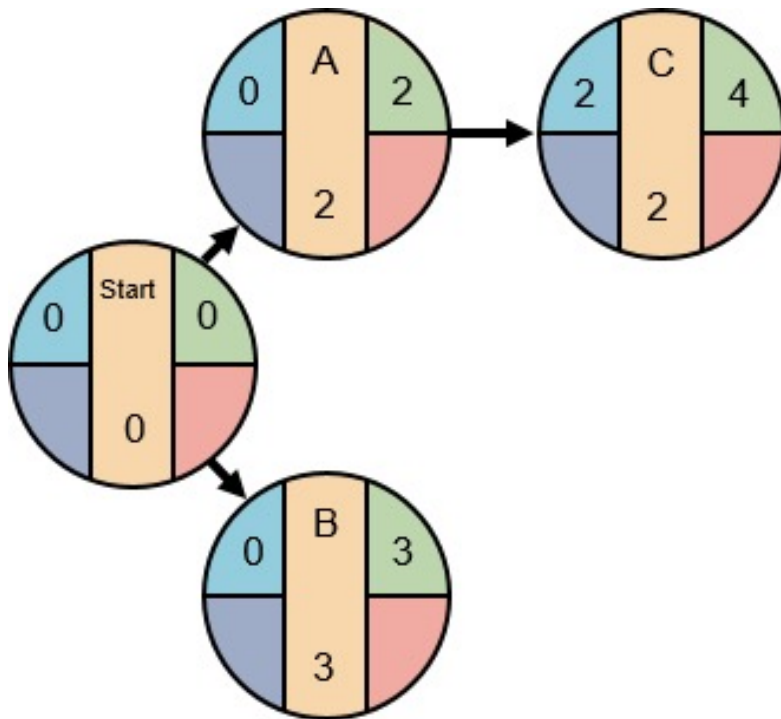
ES/EF Network



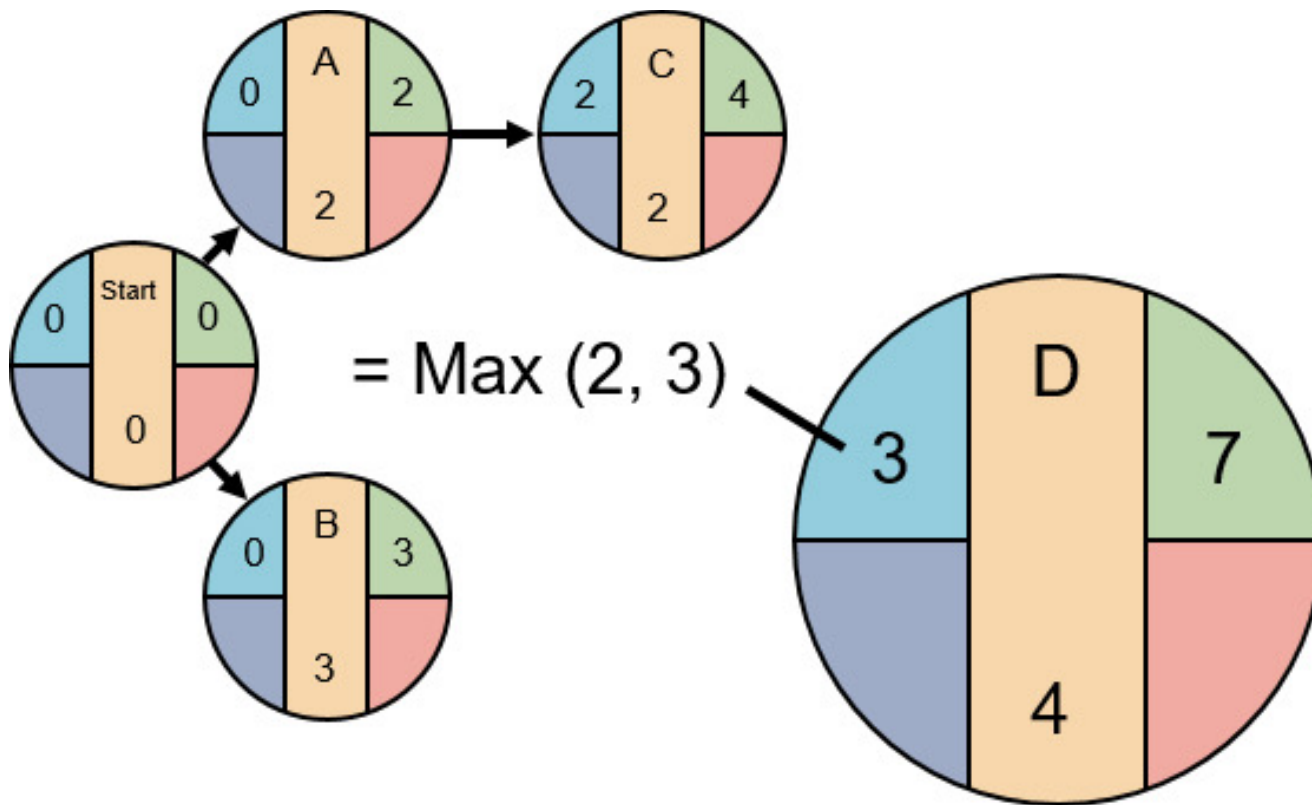
ES/EF Network



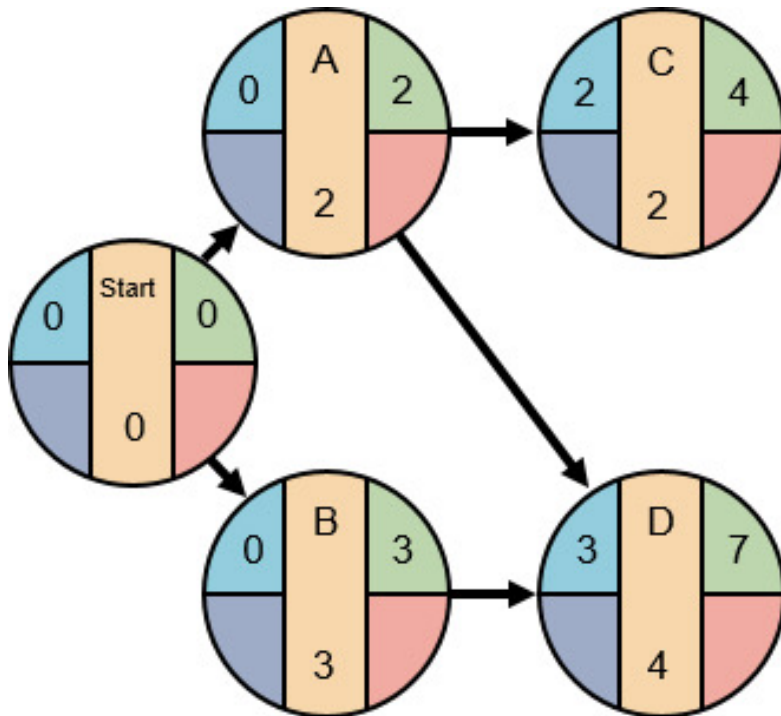
ES/EF Network



ES/EF Network

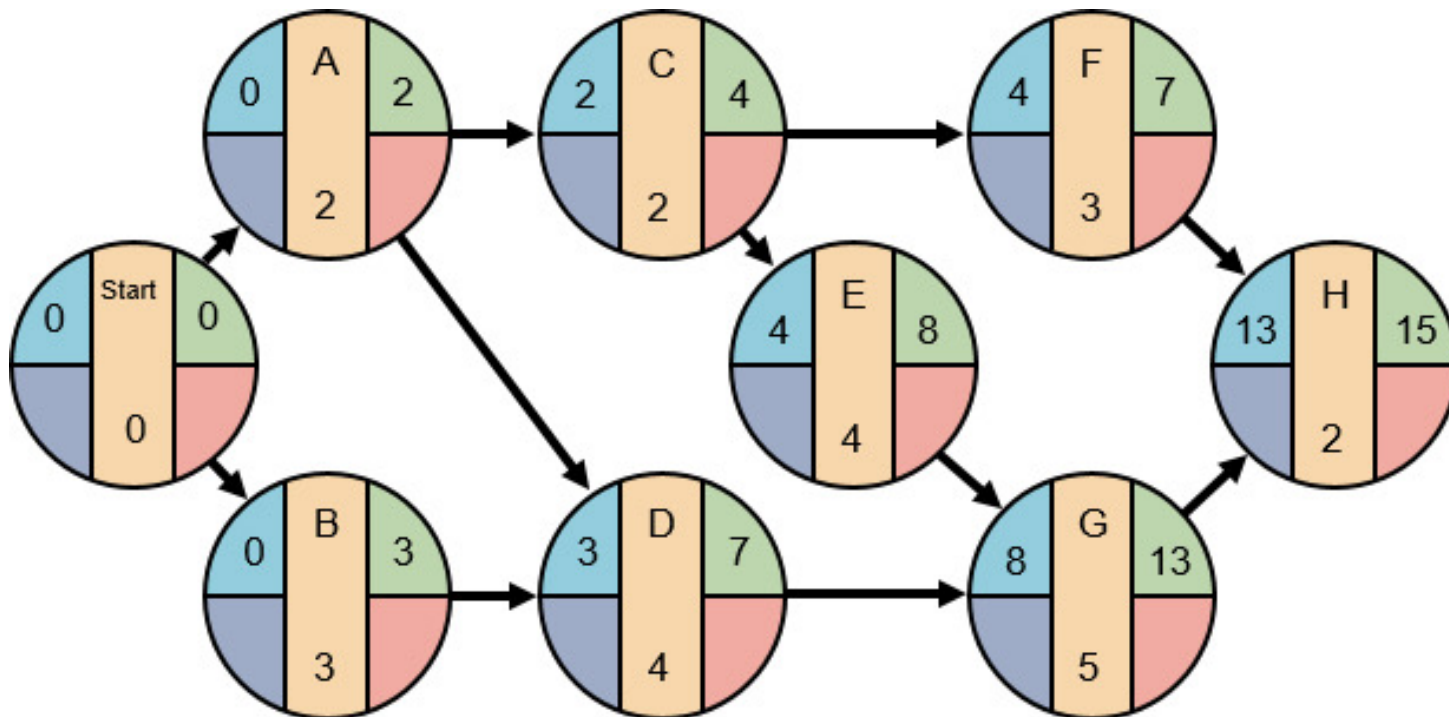


ES/EF Network



ES/EF Network

Earliest Start and Earliest Finish Times



Backward Pass

Begin with the last event and work backwards

Latest Finish Time Rule:

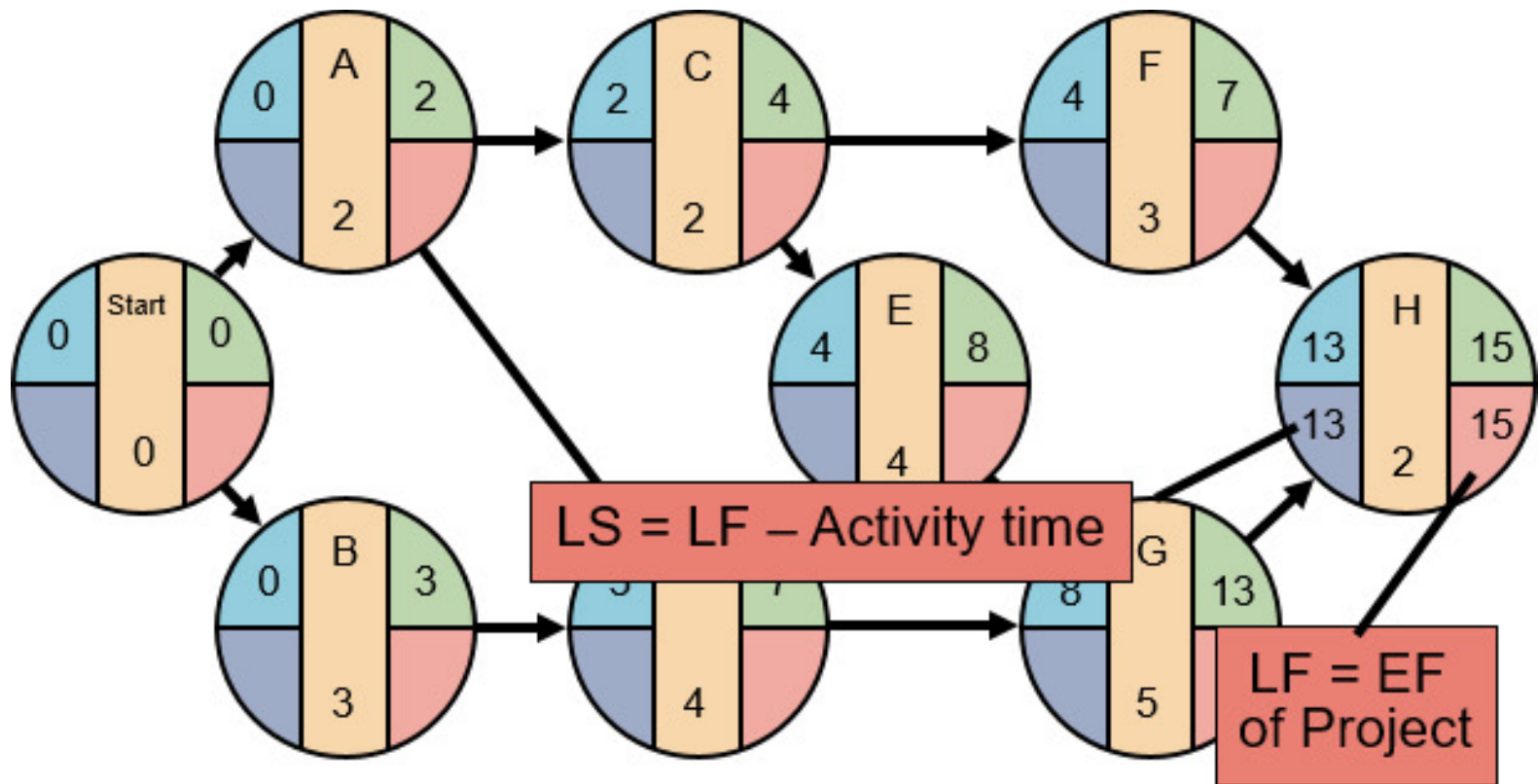
- If an activity is an immediate predecessor for just a single activity, its LF equals the LS of the activity that immediately follows it
- If an activity is an immediate predecessor to more than one activity, its LF is the minimum of all LS values of all activities that immediately follow it

$$LF = \text{Min} \{LS \text{ of all immediate following activities}\}$$

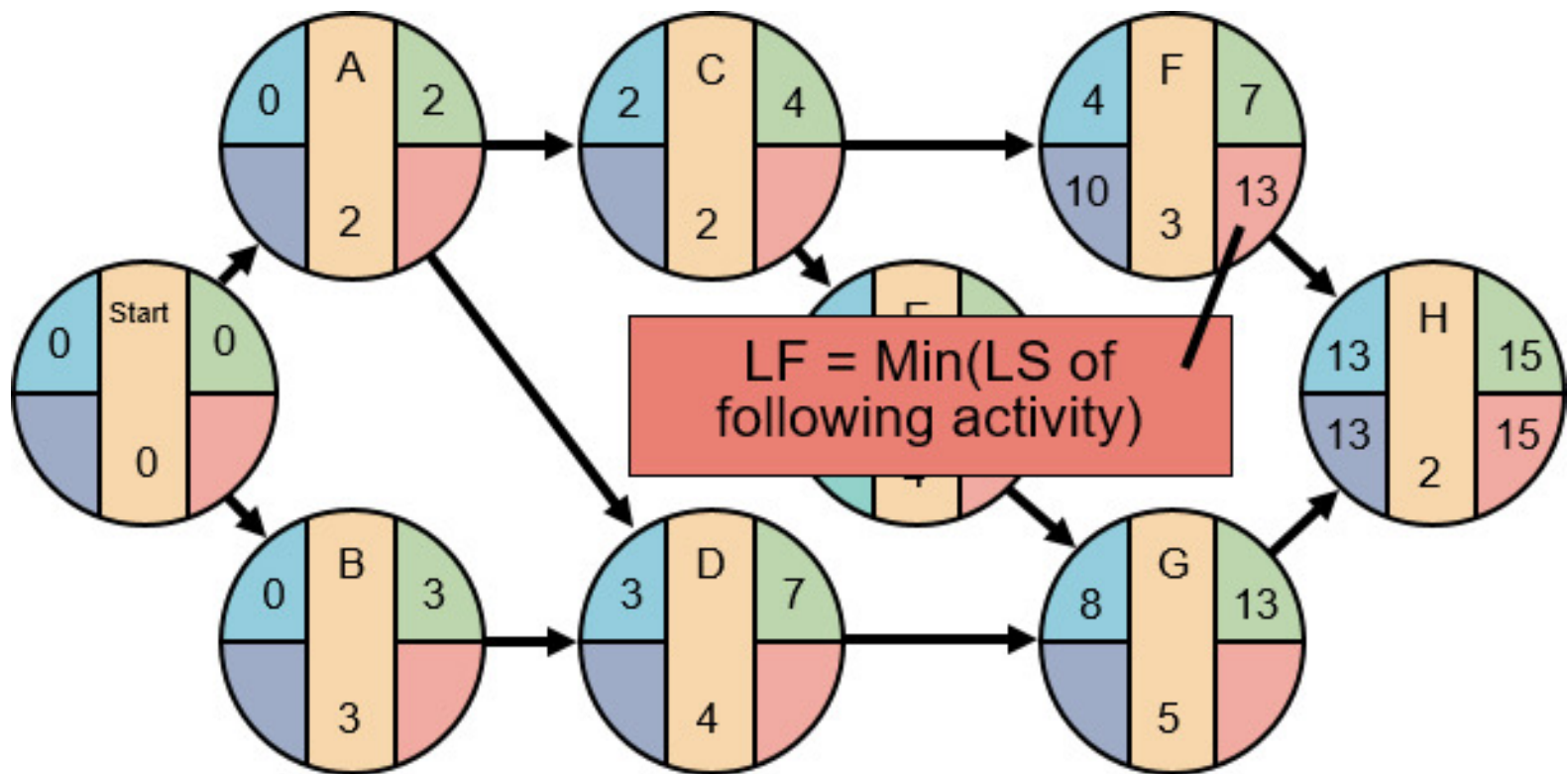
- The latest start time (LS) of an activity is the difference of its latest finish time (LF) and its activity time

$$LS = LF - \text{Activity time}$$

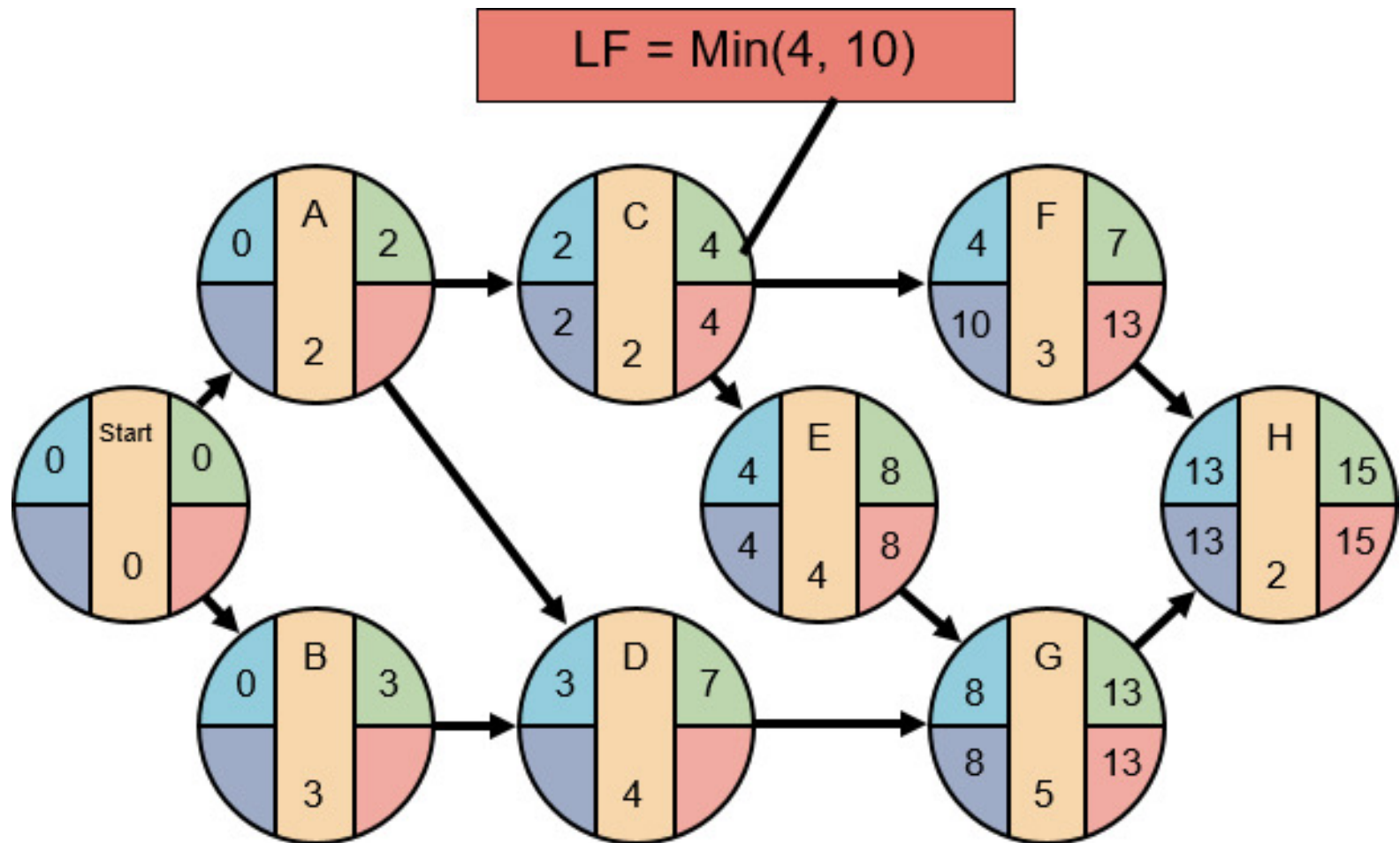
LS/LF Times



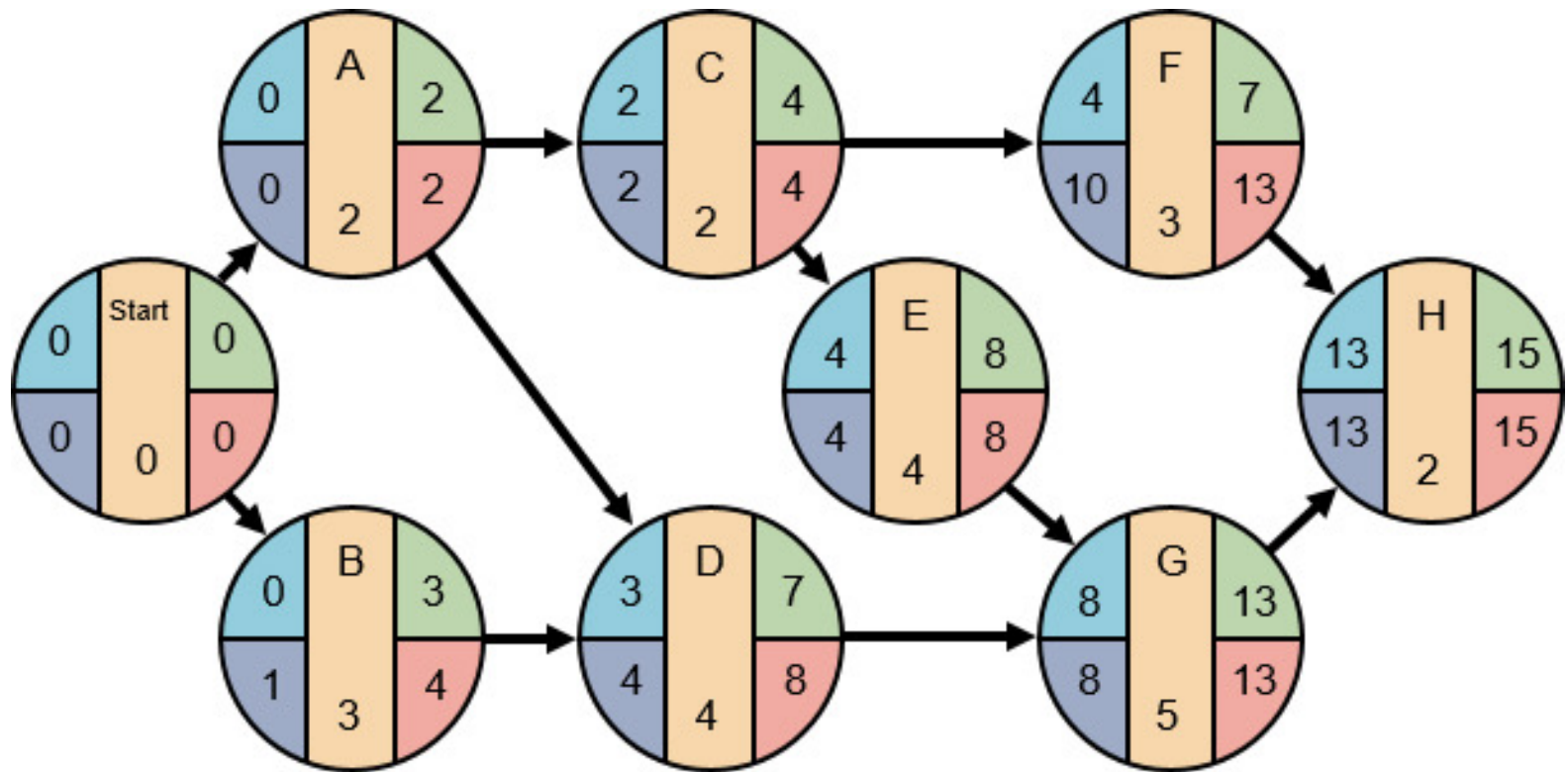
LS/LF Times



LS/LF Times



LS/LF Times



Computing Slack Time

After computing the ES, EF, LS, and LF times for all activities, compute the slack or free time for each activity

- Slack is the length of time an activity can be delayed without delaying the entire project

$$\text{Slack} = \text{LS} - \text{ES} \text{ or } \text{Slack} = \text{LF} - \text{EF}$$

Computing Slack Time

Paper manufacturing Schedule and Slack Times

Activity	Earliest Start ES	Earliest Finish EF	Latest Start LS	Latest Finish L F	Slack LS – ES	On Critical path
A	0	2	0	2	0	Yes
B	0	3	1	4	1	No
C	2	4	2	4	0	Yes
D	3	7	4	8	1	No
E	4	8	4	8	0	Yes
F	4	7	10	13	6	No
G	8	13	8	13	0	Yes
H	13	15	13	15	0	Yes

Computing Slack Time

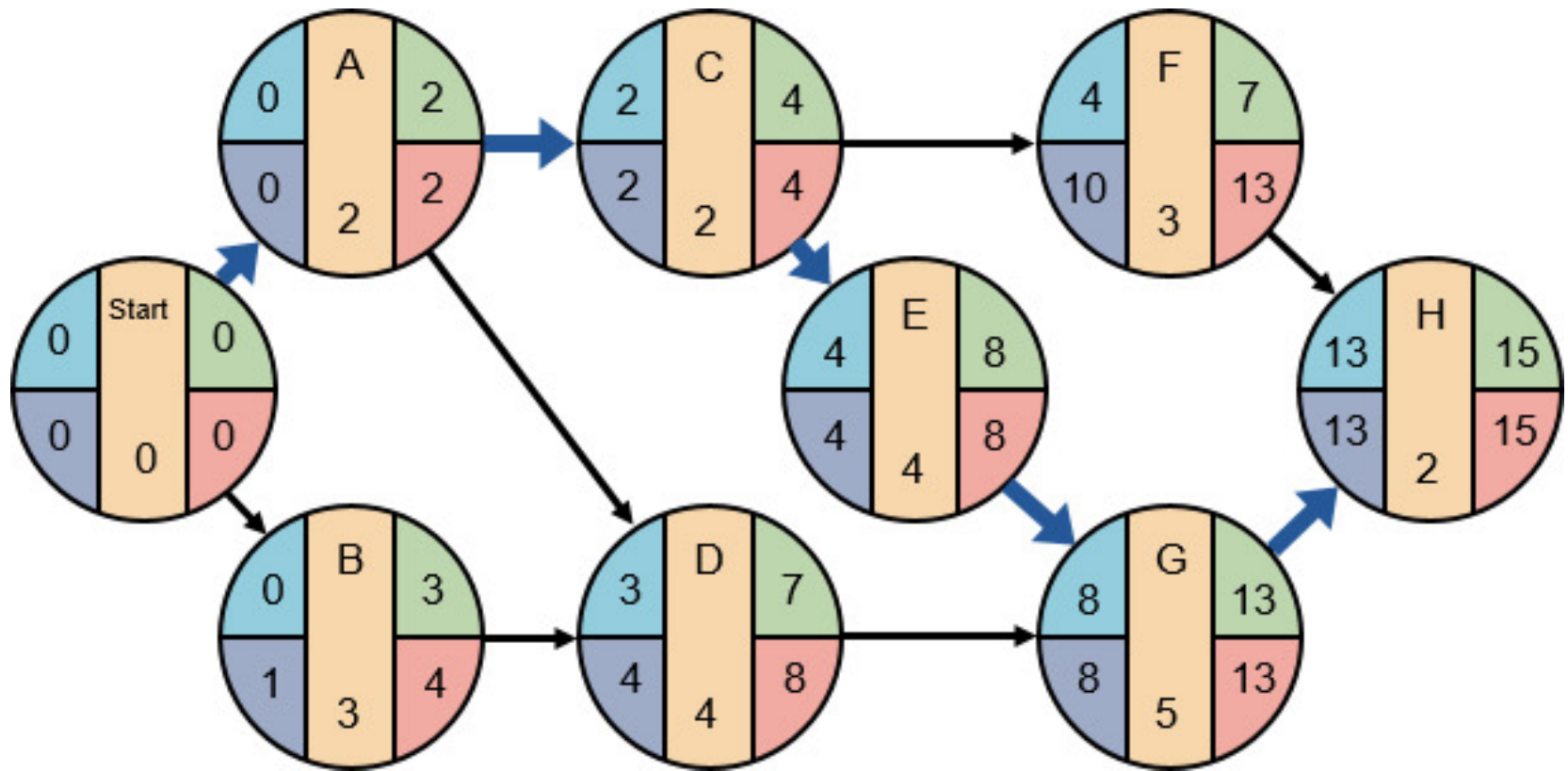
Table [continued]

Activity	Earliest Start ES	Earliest Finish EF	Latest Start LS	Latest Finish LF	Slack LS – ES	On Critical path
A	0	2	0	2	0 →	Yes
B	0	3	1	4	1	No
C	2	4	2	4	0 →	Yes
D	3	7	4	8	1	No
E	4	8	4	8	0 →	Yes
F	4	7	10	13	6	No
G	8	13	8	13	0 →	Yes
H	13	15	13	15	0 →	Yes

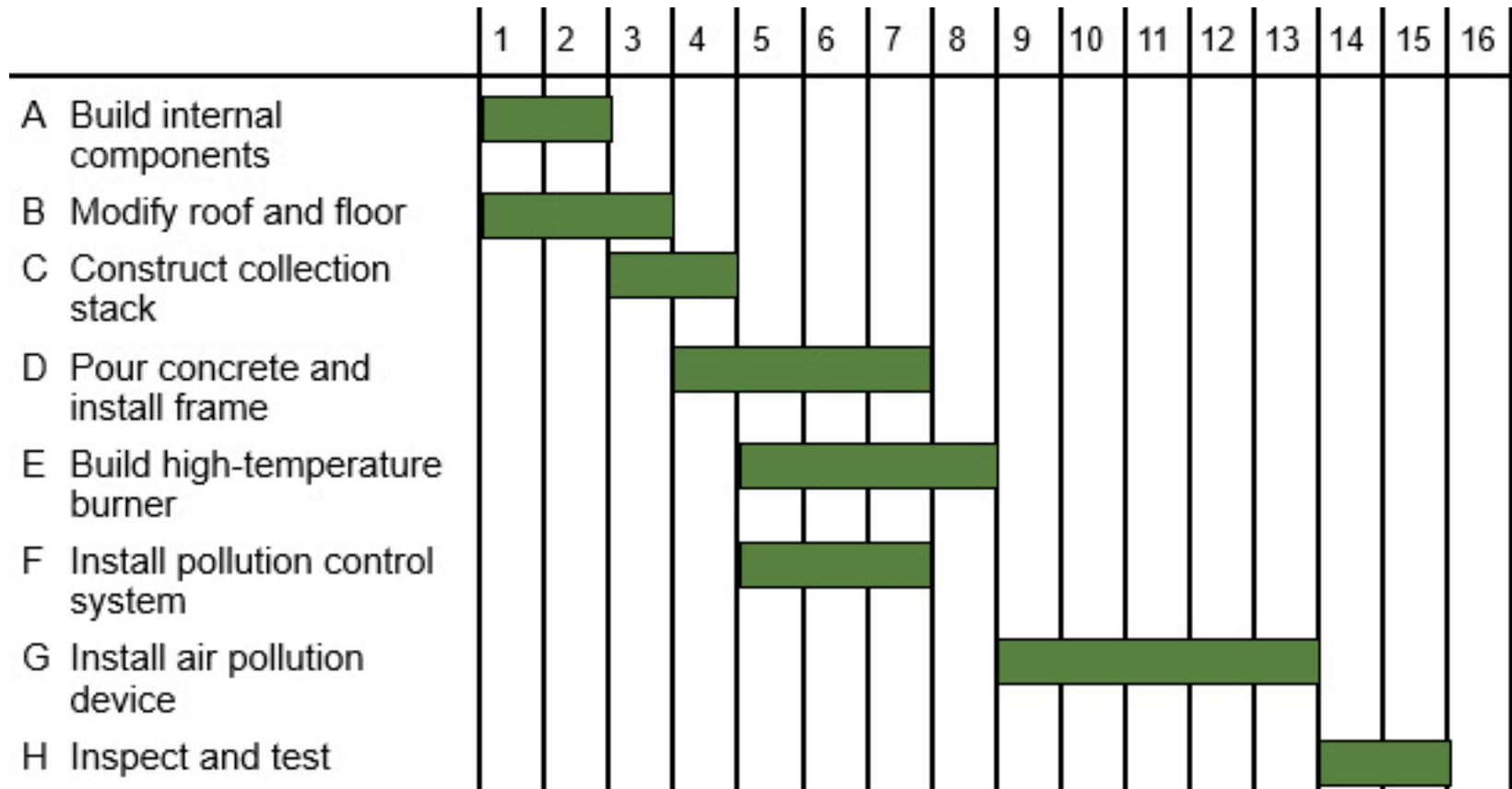
Computing Slack Time

- Activities with zero slack are on the critical path. It:
 - Starts at the first activity in the project
 - Terminates at the last activity in the project
 - Includes only critical activities

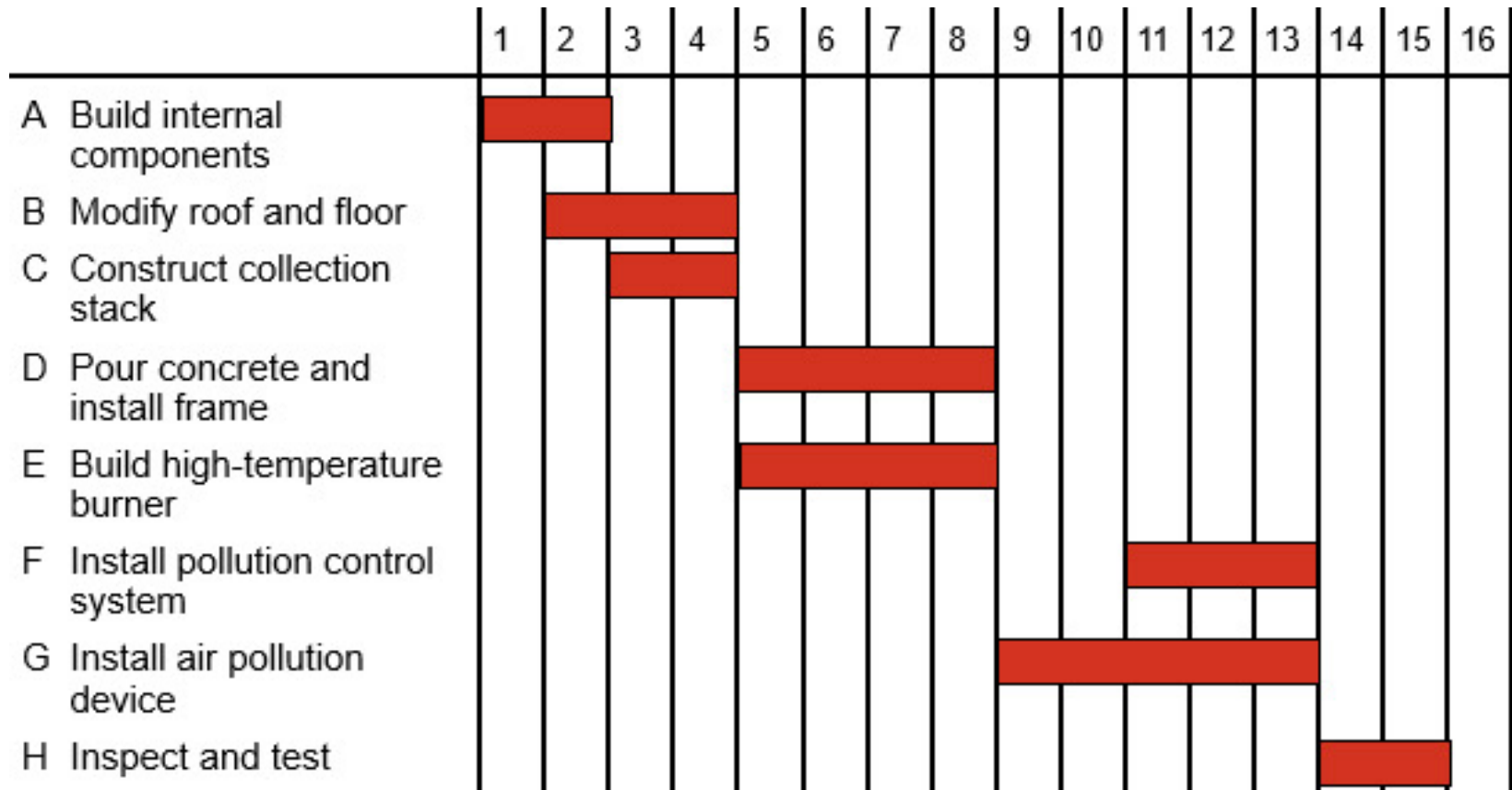
Critical Path for Paper Manufacturing



ES – EF Gantt Chart



LS – LF Gantt Chart for Milwaukee Paper



Variability in Activity Times

- CPM assumes we know a fixed time estimate for each activity and there is no variability in activity times
- PERT uses a probability distribution for activity times to allow for variability

Variability in Activity Times

- Three time estimates are required
 - **Optimistic time (a)** – if everything goes according to plan
 - **Pessimistic time (b)** – assuming very unfavorable conditions
 - **Most likely time (m)** – most realistic estimate

Variability in Activity Times

Estimate follows beta distribution

Expected activity time:

$$\frac{t = (a + 4m + b)}{6}$$

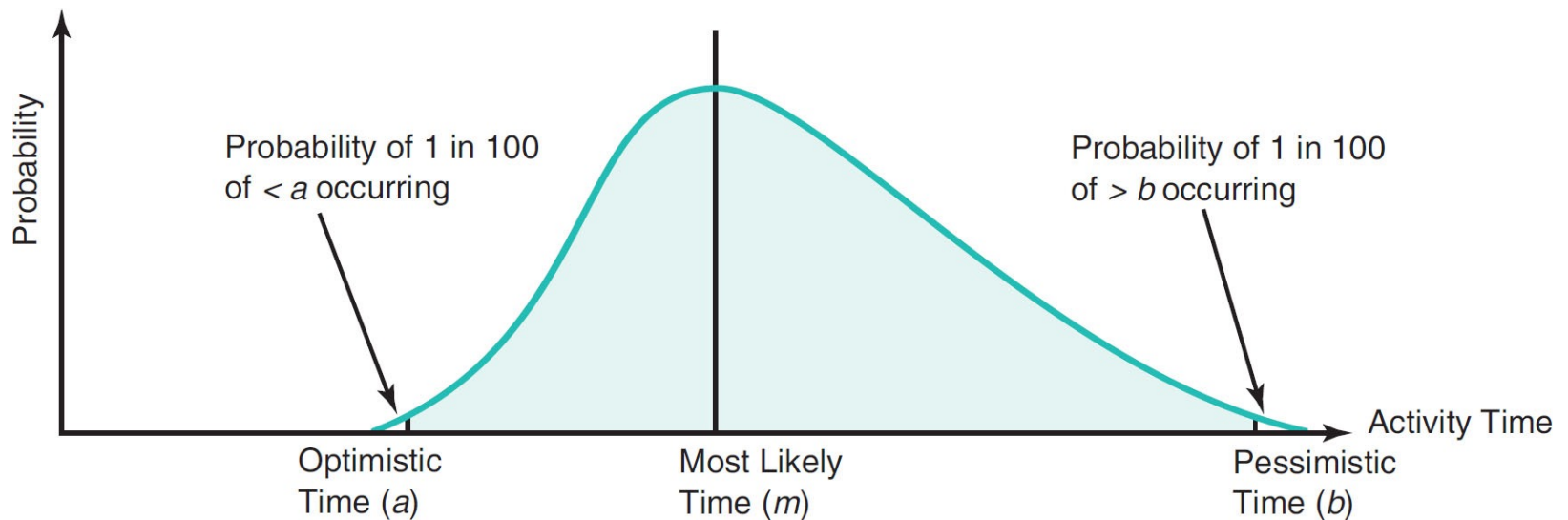
Variance of activity completion times:

$$v = \left[\frac{(b - a)}{6} \right]^2$$

Variability in Activity Times

Estimate follows beta distribution

Beta Probability Distribution with Three Time Estimates



Computing Variance

Time Estimates (in weeks) for Paper manufacturing project

Activity	Optimistic <i>a</i>	Most Likely <i>m</i>	Pessimistic <i>b</i>	Expected Time $t = \frac{(a + 4m + b)}{6}$	Variance $\left[\frac{(b - a)}{6} \right]^2$
A	1	2	3	2	.11
B	2	3	4	3	.11
C	1	2	3	2	.11
D	2	4	6	4	.44
E	1	4	7	4	1.00
F	1	2	9	3	1.78
G	3	4	11	5	1.78
H	1	2	3	2	.11

Probability of Project Completion

Project variance is computed by summing the variances of critical activities

$$\begin{aligned}\sigma_p^2 &= \text{Project variance} \\ &= \sum (\text{variances of activities on critical path})\end{aligned}$$

Probability of Project Completion

Project variance

$$\sigma_p^2 = .11 + .11 + 1.00 + 1.78 + .11 = 3.11$$

Project standard deviation

$$\begin{aligned}\sigma_p &= \sqrt{\text{Project variance}} \\ &= \sqrt{3.11} = 1.76 \text{ Weeks}\end{aligned}$$

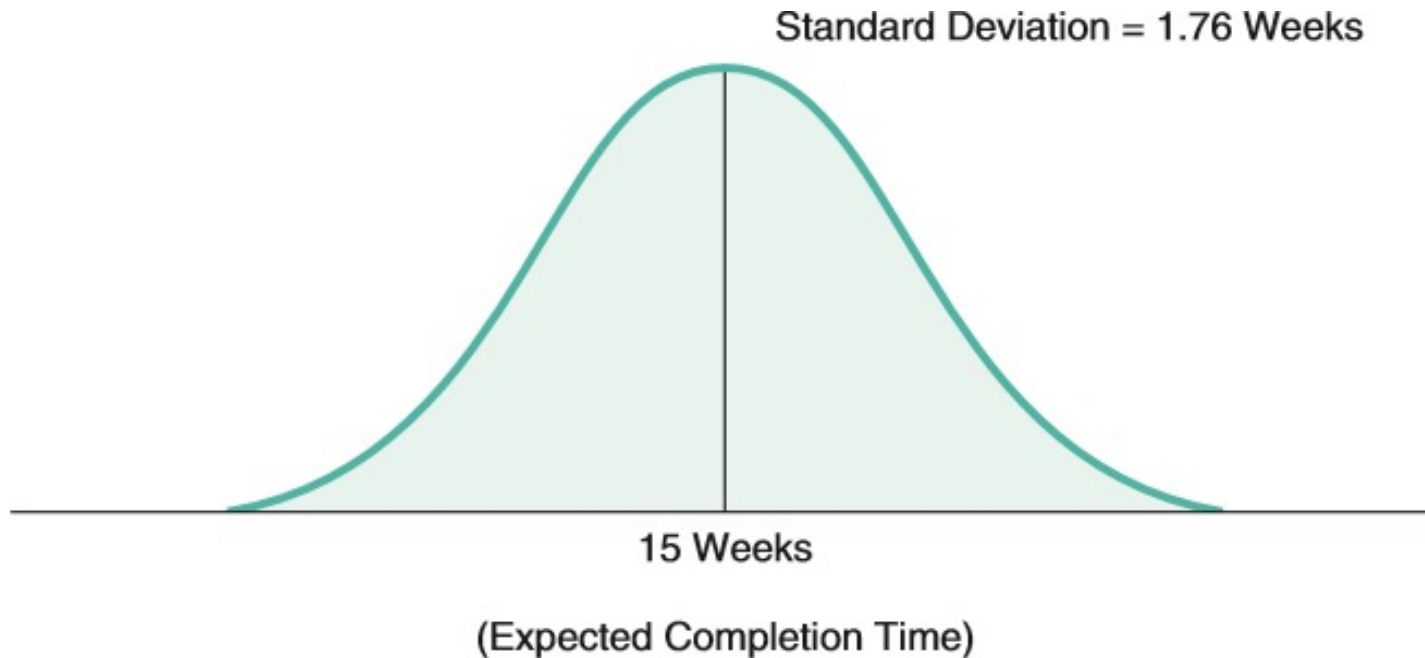
Probability of Project Completion

PERT makes two more assumptions:

- Total project completion times follow a normal probability distribution
- Activity times are statistically independent

Probability of Project Completion

Probability Distribution for Project Completion Times for Paper manufacturing project



Probability of Project Completion

What is the probability this project can be completed on or before the 16 week deadline?

$$\begin{aligned} Z &= \frac{[\text{Due date} - \text{Expected date of completion}]}{\sigma_p} \\ &= \frac{(16 \text{ weeks} - 15 \text{ weeks})}{1.76} \\ &= 0.57 \end{aligned}$$

Where Z is the number of standard deviations the due date or target date lies from the mean or expected date

Probability of Project Completion

From Appendix I

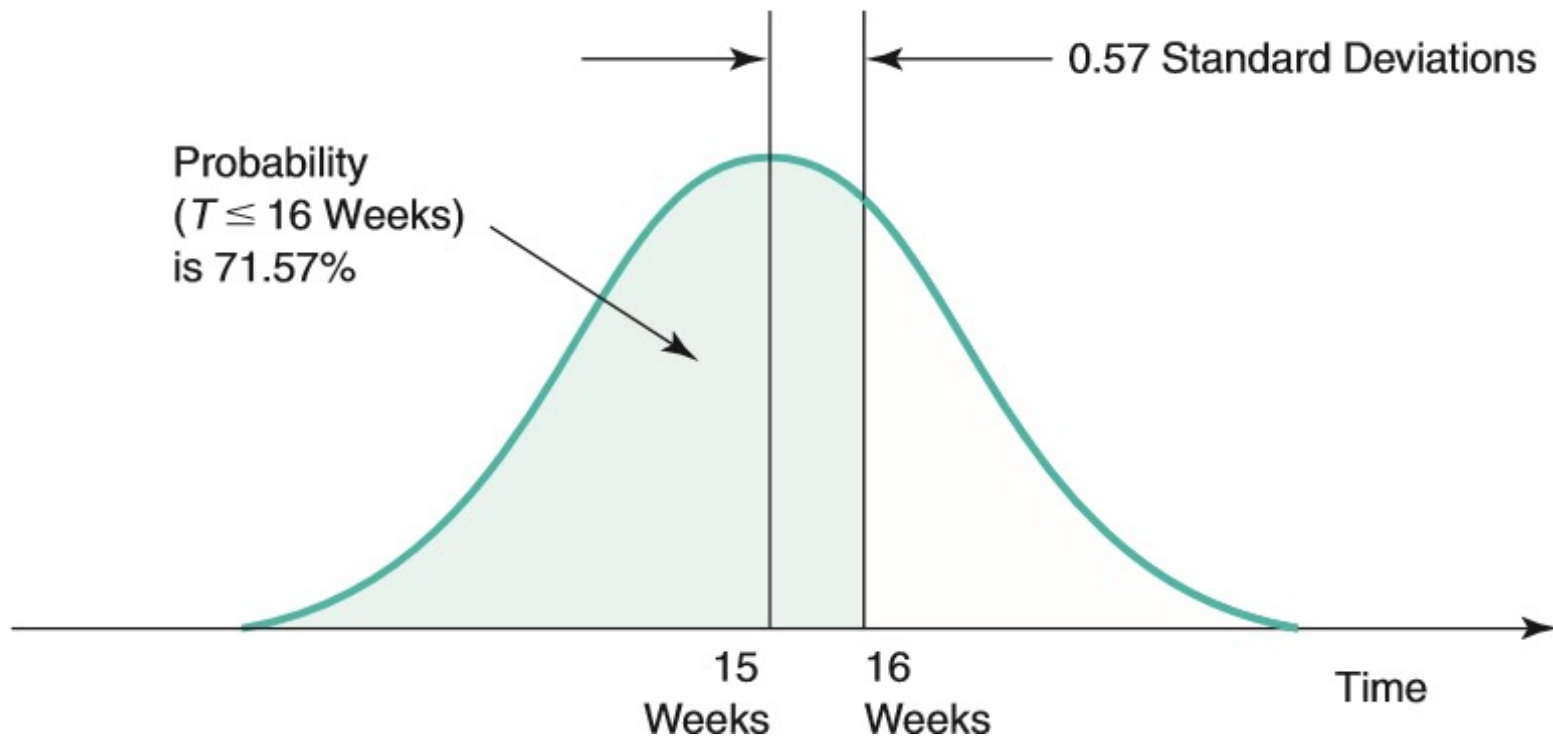
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.1	.50000	.50399		.52790	.53188
.2	.53983	.54380		.56749	.57142
≈					
.5	.69146	.69497		.71566	.71904
.6	.72575	.72907		.74857	.75175

= 0.57

Where Z is the number of standard deviations the due date or target date lies from the mean or expected date

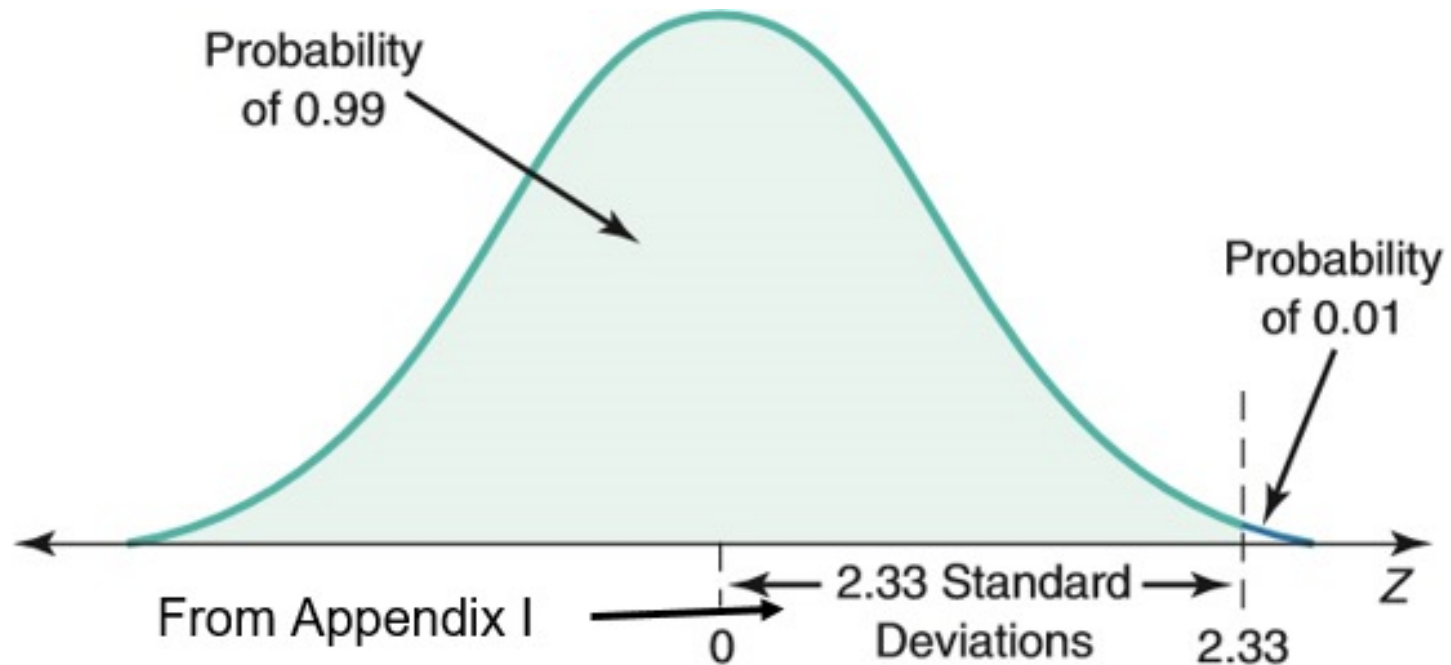
Probability of Project Completion

Probability that project will meet the 16-Week Deadline



Determining Project Completion Time

Z -Value for 99% Probability of Project Completion



Activity	Predecessor(s)	Duration (weeks)		
		<i>a</i>	<i>m</i>	<i>b</i>
A	—	6	7	8
B	—	1	2	9
C	—	1	4	7
D	A	1	2	3
E	A, B	1	2	9
F	C	1	5	9
G	C	2	2	8
H	E, F	4	4	4
I	E, F	4	4	10
J	D, H	2	5	14
K	I, G	2	2	8

- Construct the project network.
- Find the expected duration and variance of each activity.
- Find the critical path and the expected project completion time.
- What is the probability of completing the project on or before 25 weeks?
- If the probability of completing the project is 0.84, find the expected project completion time.

Variability of Completion Time for Noncritical Paths

- Variability of times for activities on noncritical paths must be considered when finding the probability of finishing in a specified time
- Variation in noncritical activity may cause change in critical path

What Project Management Has Provided So Far

1. The project's expected completion time is 15 weeks
2. There is a 71.57% chance the equipment will be in place by the 16 week deadline
3. Five activities (A, C, E, G, and H) are on the critical path
4. Three activities (B, D, F) are not on the critical path and have slack time
5. A detailed schedule is available

Advantages of P E R T/C P M

1. Especially useful when scheduling and controlling large projects
2. Straightforward concept and not mathematically complex
3. Graphical networks help highlight relationships among project activities
4. Critical path and slack time analyses help pinpoint activities that need to be closely watched
5. Project documentation and graphics point out who is responsible for various activities
6. Applicable to a wide variety of projects
7. Useful in monitoring not only schedules but costs as well

Limitations of PERT/CPM

1. Project activities have to be clearly defined, independent, and stable in their relationships
2. Precedence relationships must be specified and networked together
3. Time estimates tend to be subjective and are subject to fudging by managers
4. There is an inherent danger of too much emphasis being placed on the longest, or critical, path

TABLE 10.10 Example 10.10

Activity	Normal time (weeks)	Normal cost (Rs.)	Crash time (weeks)	Crash cost (Rs.)
1-2	13	700	9	900
1-3	5	400	4	460
1-4	7	600	4	810
2-5	12	800	11	865
3-2	6	900	4	1130
3-4	5	1000	3	1180
4-5	9	1500	6	1800

If the indirect cost per week is Rs. 160, find the optimal crashed project completion time.